LARA: overall training architecture

DiT backbone

Sample layers → predict / compare → loss

01 Flow matching

v_θ

L_{FM}

Velocity target

u_t

LARA objectives

Hidden layers

02 Prediction

Random pair (a, b)

A_a

A_b

Gψ

$\hat{A}_{a\to b}$

$e_t + e_a + e_b$

1 random pair per training iteration

L_{pred}

sg

03 Transition consistency · a < b < c

sg(A_a)

Gψ : a → c

cached prediction

$\hat{A}_{a\to b}$

sg

EMA : b → c

sg

L_{cons}

04 Redundancy

Random pair (i, j)

Sampled independently of (a, b)

A_i

A_j

1 random pair per training iteration

L_{div}

x_t

Shared transition predictor Gψ

e_t + e_a + e_b

Embeddings

e_t timestep

e_a source layer

e_b target layer

AdaLN / FiLM

Condition on timestep + source / target depths

A_a

Linear 768 → 384

Spatial blocks × 10

Layer norm

Linear 384 → 768

$\hat{A}_{a\to b}$

16 × 16 tokens

Conditioned spatial block

Layer norm

γ, β

Scale + shift

ConvNeXt

residual

LN · FiLM · MHSA

+ projection + residual

after blocks 4 and 8

Block output

$L = L_{FM} + L_{LARA}$